

Rutvi Khamar

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Education

Florida Institute of Technology

M.S. in Computer Science (GPA: 3.93 / 4.00)

Aug 2023 - May 2025

Melbourne, Florida

Relevant Coursework: Applied Discrete Mathematics, Machine Learning (coded algorithms from scratch), Big Data (Spark/Hadoop/AWS), NLP, Thesis (Early breast cancer detection using Non-negative Matrix Factorization)

Adani University

B.Tech in Information and Communication Technology (GPA: 8.93 / 10.00)

Aug 2019 - May 2023

Ahmedabad, Gujarat

Relevant Coursework: Data structures & algorithms, Object oriented programming, Data Mining & BI, Linear Algebra, Probability & Statistics, Advanced Software Engineering

Technical Skills

Programming: Python, SQL, Java

Frameworks: Numpy, Pandas, PyTorch, Tensorflow, Transformers, HuggingFace, Transformers, Dask, spaCy, Prodigy, PySpark, LangChain, LangGraph, Claude Agent SDK, MLflow, Weights & Biases, scikitlearn, matplotlib, plotly, Streamlit

Certification: Technical AI Safety (BlueDot Impact) – Covered RLHF, constitutional AI, behavioral evaluations, interpretability, and harm minimization techniques; studied safety frameworks from Anthropic, OpenAI, and DeepMind

Experience

Steppingblocks | AI/ML Engineer - Melbourne, FL

Aug 2025 – Present

- Built a job title to SOC codes classification pipeline using RAG, cross-encoders, and a domain-specific embedding model fine-tuned on resume and skills data – improving semantic clustering for related concepts general models miss
- Built an embedding visualization tool to empirically compare general vs. domain-specific models, demonstrating tighter clustering of semantically related job titles and skills – used to validate and justify model selection
- Engineered a 150K-skill definition generation pipeline using Groq's inference API and batch processing, solving cost and latency bottlenecks that made hosted APIs prohibitive and local models unstable at scale – structured prompts processed 10 skills per call with JSON output capturing skill type, enabling reliable parsing
- Rebuilt a record linkage workflow crashing on large datasets by migrating to Spark, while redesigning blocking, training, and comparison functions to improve entity matching accuracy from ~60% to ~80%
- Building a multi-agent system to automate the record linkage process – including a text-to-SQL agent with sandboxed code execution, an automated data preprocessing agent, and an orchestration layer coordinating agents end-to-end with minimal human intervention

L3Harris Institute for Assured Information | Research Assistant - Melbourne, FL

May 2024 – July 2024

- Conducted systematic research comparing multiple entity extraction approaches (SpaCy+ClausalE, Stanza-CoreNLP, fine-tuned LLaMA2, T5-based transformers) to identify optimal methods for converting unstructured text into structured knowledge graphs, evaluating trade-offs between accuracy, computational cost, and scalability.
- Designed and executed experiments for fine-tuning LLaMA2 for structured query generation with limited compute resources, implementing prompt engineering and parameter-efficient fine-tuning strategies to enable text-to-Cypher translation for graph database querying.
- Built end-to-end GraphRAG research prototype integrating LLMs, Neo4j graph database, to explore how graph-structured knowledge enhances LLM reasoning over complex relational data.
- Documented comparative analysis across all approaches, evaluating each method's strengths and failure modes to guide future research directions.

Projects

GraphRAG for Healthcare | *Neo4j, LLMs, Cypher, Prompt Engineering, Entity Extraction* 🔗

- Built a GraphRAG pipeline on the Semantic MEDLINE Database (SemMedDB) to reduce LLM hallucinations in healthcare — constructing a knowledge graph in Neo4j and enabling natural language querying via LLM-generated Cypher using prompt engineering and entity extraction.
- Demonstrated that graph-structured retrieval captures higher-order entity relationships missed by standard RAG, improving contextual accuracy of LLM responses in medical Q&A.

FlightMate | *LangChain, ArangoDB, AQL, NetworkX, Gradio* 🔗

- Built a multi-tool agentic chatbot over a graph database of airports, flights, and airlines — dynamically routing queries across tools for shortest-path computation, live weather lookup, and airline statistics.
- Designed a hybrid tool combining AQL graph queries with NetworkX-based algorithms and interactive visualizations, enabling the agent to answer complex relational queries with real-time graph rendering.

High School Helper – AI Agent | *LangGraph, Groq Cloud, OpenAI OSS-120B, SymPy, Streamlit* 🔗

- Built a ReAct agent using LangGraph that solves math and physics problems in plain English — combining LLM reasoning with a SymPy-based sandboxed code execution tool for exact symbolic and mathematical computation.
- Eliminates LLM hallucinations in numerical reasoning by grounding all calculations in symbolic execution rather than model-generated approximations.

Research Work

Early breast cancer detection with 3D ultrasound using NMF (Thesis) | *Advisor: Dr. Mitra*

- Published and presented at SPIE Medical Imaging Conference 2025 — a novel ML approach for early-stage breast cancer detection using Non-negative Matrix Factorization on 3D ultrasound data.
- Developed and optimized multiple NMF variants with custom regularization to extract diagnostic features from DCE-US imaging data, distinguishing benign from malignant lesions without contrast agents.
- Achieved 75.5% sensitivity and 78.3% specificity using only pre-injection data — matching contrast-enhanced performance while eliminating patient risk, demonstrating that smarter feature extraction can replace invasive procedures.